

Griffin Burke

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EDUCATION

University of Illinois Urbana-Champaign

August 2023 – May 2027

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 4.00

Coursework: Database Systems, System Programming, Algorithms and Models of Computation, Data Structures, Numerical Analysis
Spring 2026: Artificial Intelligence, Deep Learning for Computer Vision, Programming Languages and Compilers

APPLIED EXPERIENCE

Illinois Digital Animal Systems Lab

Urbana, IL

Undergraduate Research Assistant

January 2025 – Present

- Developing a system of YOLOv11 computer vision models to detect and segment overhead cattle images, extract relevant features, and utilize them to predict cattle weight and body condition score
- Achieved a test mean average precision of 0.994 in segmenting relevant portions of overhead cattle images
- Recovered and reconstructed a 200+ GB ROS bag file with corrupted frames, preserving over 350,000 training images

Kashmir World Foundation

Greenwood, AR (Remote)

AI Team Lead

May 2025 – Present

- Lead development of machine learning models for the detection of poaching and other illegal activity in the rainforest
- Coordinate across multiple teams to develop AI deployed on the nodes of a mesh network that detects human presence
- Led multiple cohorts of interns on the data analysis and modeling of 500+ hours of rainforest audio and environmental data

Machine Learning Engineer Intern

September 2024 – May 2025

- Conducted initial training and analysis on the efficiency of different sized YOLOv8 models for detecting rainforest species through trail cams, resulting in a lightweight model with a precision and recall of 88.5% and 88.2%
- Fine-tuned the optimal base model with systematic hyperparameter sweeps and training data augmentation, resulting in a 3.5% increase in precision and recall

LEADERSHIP & INVOLVEMENT

Course Assistant

Probability and Statistics for Computer Science (CS361)

August 2025 – Present

- Support course operations through grading homework and holding office hours, teaching probability distributions, maximum likelihood estimation, principal component analysis, linear regression, Naïve Bayes, support vector machines, Markov chains

Linear Algebra with Computational Applications (MATH257)

January 2025 – August 2025

- Assisted with in-person Python-based lab sections and held online office hours for homework questions, teaching concepts such as linear transformations, least squares, linear regression, singular value decomposition, principal component analysis

Project Code UIUC

Urbana, IL

Vice President

May 2025 – Present

- Oversee student organization logistics, mentor project managers, and provide support for other leadership duties

Project Manager

September 2024 – May 2025

- Led a team of 14 on the development of CompanyMatch (a website leveraging sentence embeddings and semantic similarity to match users to companies) across three sub teams for two semesters

Developer

February 2024 – May 2024

- Collaborated with a team of 12 to build the UIUC Involvement Bot, contributing heavily to API and ML development

PERSONAL PROJECTS

Note Transfer | Python, TypeScript, FastAPI, Svelte, TailwindCSS

May 2025 – Present

- Building an AI system using YOLOv11 vision and Microsoft TrOCR models to convert handwritten notes into markdown
- Architected a page-parsing system utilizing computer vision at multiple levels, converting notes into a by-line document tree

UIUC Involvement Bot | Python, Flask, Keras, HTML/CSS/JS, SQLite

February 2024 – June 2024

- Developed, trained, and tested two neural network models with Keras, achieving accuracies of 89.4% and 95.6%
- Contributed to the end-to-end development of a club recommendation website for UIUC students, using neural networks for classification of the user's preferred club category and intensity level

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, SQL, HTML/CSS, Kotlin, Java

Libraries: PyTorch, Scikit-learn, pandas, numpy, Ultralytics, Keras

Tools and Frameworks: Flask, FastAPI, React, Svelte, Express, SQLite, Git, Google Cloud Platform